SWEDEN

AI IN THE HEALTHCARE SECTOR

CISCO Field Project

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E-HEALTH VISION STATEMENT

To ensure a secure, open, and resilient digital society in Sweden, where cybersecurity supports democratic values, innovation, and national interests.

*"In 2025, Sweden will be best in the world at using the opportunities offered by **digitisation** and **eHealth** to make it easier for people to achieve good and **equal health** and **welfare**, and to develop and strengthen their own **resources** for increased **independence** and participation in the life of society."*

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DIGITAL PRIORITIES

- Universal Digital Access and Inclusion
- Economic Growth via Digital Innovation
- Democracy in the Digital Realm
- International Cyber Collaboration
- Emerging Technology in Security Policy

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CYBERSECURITY PRIORITIES

1. Clear **division of responsibilities** across public and private sectors.
2. **Crisis management structures** tailored to cyber incidents.
3. **Legal and regulatory frameworks** for better security enforcement.
4. **Protection of critical infrastructure** in the cyber realm.
5. **Supply chain cybersecurity** for IT products and services.
6. **Public-private cooperation** and secure digital services for citizens.

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AI IN HEALTHCARE

CURRENT USE

- **Diagnostics**: AI for radiology, skin cancer detection, early diagnosis.
- **Hospital efficiency**: AI for patient flow and clinical performance tracking.
- **Emergency rooms**: AI-assisted triage for faster/accurate symptom assessments.
- **AI-powered chatbots**: Early support systems for patient engagement and initial diagnostics.

PLANNED USE

- AI-powered **personalized medicine** (via genomic data).
- **Deeper integration** with eHealth strategies to support a resilient, secure digital health ecosystem.
- Further deployment in **operational logistics** to optimize resources and costs.

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STAKEHOLDER ANALYSIS

STAKEHOLDERS' NEEDS, INTERESTS & CONCERNS

- **Data protection**
- **Ease of access** + **equality of access** (e.g. no algorithmic bias)
- **Ethical implications**
- **Transparency**
- Workforce **training** and readiness

- **Improved** healthcare outcomes and **efficiency**
- **Accountability** in AI procurement and implementation
- **Job security**
- **Cost-effectiveness** & **budget constraints**
- **Social welfare**

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WHAT IS AT STAKE?

ECONOMIC

- **Financial loss** from ransomware, litigation and IT recovery
- **Operational downtime** disrupts care and revenue flow
- **Supply chain vulnerabilities**, one weak link can halt operations

SOCIETAL

- **Algorithmic bias** risks exacerbating health disparities
- **Loss of public trust** from data breaches or unexplained AI decisions
- Spread of **misinformation** or **AI misuse** can undermine patient safety and care-seeking behaviour

SECURITY

- **Hacking** and **sabotage** of AI-powered critical systems (e.g., diagnostics)
- **Data breaches** affecting privacy and potentially national security
- **Threats to availability of care**: cyberattacks can endanger lives by disabling emergency services
- **Hybrid threats** combining cyberattacks with disinformation campaigns

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POLICY SCOPE

- TRANSPARENCY & ACCOUNTABILITY
- FAIRNESS & INCLUSION
- EFFECTIVE DECISION-MAKING
- PRIVACY & DATA PROTECTION

This policy applies to the selection and implementation of AI tools and services in Sweden's healthcare sector in order to achieve accountability & transparency, fairness & inclusion, effectiveness in decision-making, and data protection in the following activities: ***personalized diagnostics and assessments, AI-assisted decision-making, data analysis & predictive analytics.***

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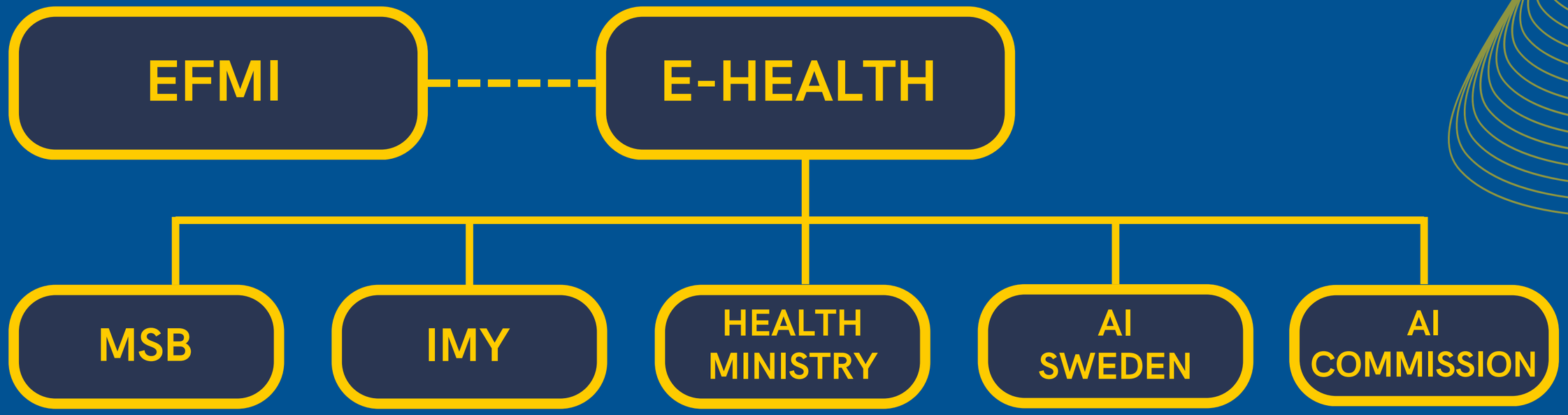
AI-HEALTH COUNCIL



GOVERNANCE BODY

E-HEALTH AGENCY

- Implements national e-health **strategies to digitalise healthcare** and social services
- Manages **electronic prescription**s and ensures secure data exchange between providers and pharmacies
- Provides **digital services** for citizens, like access to prescription data and cost protections

MSB

SWEDISH CIVIL CONTINGENCIES AGENCY

- Has branches regarding **Information & Cybersecurity** and **Crisis Management**
- **No equivalent of ACN in Sweden**. MSB is the closest agency that exists
- Provides **national oversight**
- The competent body for **cybersecurity interest and governance**

IMY

SWEDISH AUTHORITY FOR PRIVACY AND PROTECTION

- Ensures **compliance with GDPR**
- Handles **complaints** and conducts **investigations** into potential data protection violations
- Issues fines and provides guidance to promote **lawful** and **responsible data handling practices**

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GOVERNANCE BODY

HEALTH MINISTRY

- Develops **national policies** for health care, public health, and social welfare
- Oversees **social insurance systems**, including pensions and disability benefits
- Coordinates with **regional** and **local authorities** to ensure effective delivery of health and social services

AI SWEDEN

NATIONAL CENTER FOR APPLIED AI

- Drives **AI adoption** across sectors
- Facilitates **collaboration** among industry, academia, and government
- Provides **infrastructure** and **training** to support AI development and use

AI COMMISSION

- National **advisory body** on AI policy and strategy
- Supports alignment with EU frameworks and boosts international competitiveness
- Guides the development of AI in **public and private sectors**

EFMI

EUROPEAN FEDERATION FOR MEDICAL INFORMATICS

- Advance **international cooperation** in medical informatics across Europe
- Promote **high standards** in the application and education of **medical informatics**
- Encourage **R&D** in the field of medical informatics

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ACTION PLAN

PUBLIC TENDER FOR THE SELECTION OF A EU AI PROVIDER

Sponsored by: Swedish Government, EFMI (EU), e-Health agency, EU funding

Estimated cost: €15-18 million

General requirements for participation in the public tender

- **European AI provider** which is in line with the EU strategy of AI independence (given the current geopolitical tensions)
- **Compliance with current and future European and EU regulations about AI** (and in general)
- **Transparency of the source code** which ensures **transparency** & **secure pipelines**
- **Training of models has been done on a diverse set of the population** (various ethnicities + genders + ages + other demographic features) which ensures **fairness**, **effectiveness** and **reduces/eliminates algorithmic bias**.
- Can deliver **audit logs** to ensure **traceability**, **explainability**, and **transparency**
- Provision of known limitations

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ACTION PLAN

DEVELOPMENT OF AN OVERALL FRAMEWORK & METHODOLOGY FOR IMPLEMENTATION BY THE AI-HEALTH COUNCIL

- **Convenient** way of implementing AI on a **regional level**
  - **Decentralised healthcare system**, with regional authorities taking major decisions regarding the financing and delivery of regional healthcare
- **Easier for smaller regions** to implement because they do not have to go out on their own to find a vendor and create their own framework and methodology
- **Minimum cybersecurity level of the usage of AI** in the Swedish healthcare sector
- Every region has **equal access to AI for their healthcare services**
- Ensures that the objectives of the policy scope are met as the **policy-makers** are the ones who would participate in the **creation of the framework and methodology**.

- **Note**: *Each hospital and healthcare facility are free to implement their own AI solutions. However, through the development of the framework and methodology by the government, it does not make sense for a hospital/region to decide to implement their own AI solution as it will be more costly, would require expertise in the domain, and overall more effort.*

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ACTION PLAN

CAPACITY BUILDING

- **Use of AI training & awareness** offered at the expense of the government + EU (EFMI)
  - Easier acceptance by the hospitals, healthcare workers, unions, etc.
  - Build cybersecurity culture
  - Less financial burden on smaller regions
- **Evaluation of the AI infrastructure and facilities** of the different regions and **support for the implementation** of any necessary infrastructure and facilities

NO HARD REGULATION

- **Overregulation concerns**: Already many current laws regarding AI on an EU level and many that are going to be implemented in the near future
- Implementation of a national regulation may create **inconsistencies** and **clashes** with the overall EU AI regulatory framework

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AI-CYBER RESILIENCE

SECURE DEVELOPMENT PIPELINES

- Ensured through **selection of the AI provider via public tender**
- Providers must meet **security-by-design standards** and EU compliance

AI THREAT MONITORING

- Managed by **AI Sweden**, with dedicated team for healthcare-related AI threats
- Monitors **global and domestic threat trends** in healthcare AI integration

JOINT RESPONSE TO GLOBAL THREATS

- Enabled through multistakeholder collaboration within the governance model
- EFMI's inclusion ensures strong alignment and EU-level coordination

PUBLIC-PRIVATE COORDINATION

- **Public tender process** fosters structured collaboration between government and private sector
- **AI Sweden's public-private composition** reinforces continuous cooperation

INCIDENT HANDLING & CRISIS MANAGEMENT

- Built into the **national framework & methodology**
- AI vendors must have **incident response capabilities**
- **Clear communication lines** between hospitals, MSB, IMY, and e-Health Agency

POLICY ALIGNMENT

- Fully aligned with **EU AI Act, GDPR, NIS2, Cyber Resilience Act (CRA)**
- Supports Sweden's **digital** and **cybersecurity strategies**
- **No new national** regulation to avoid overregulation and lengthy approval processes

INFORMATION SHARING

- Integrated into the national framework via direct communication between **MSB, AI Sweden, ENISA, and healthcare stakeholders**
- **Governance model** enables seamless exchange between national agencies and EU-level partners

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THANK YOU.

Q&A

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